BIOS 731 Final Presentation

BGW-xWAS Tool with application to transcriptomic and proteomic data of Alzheimer's disease dementia

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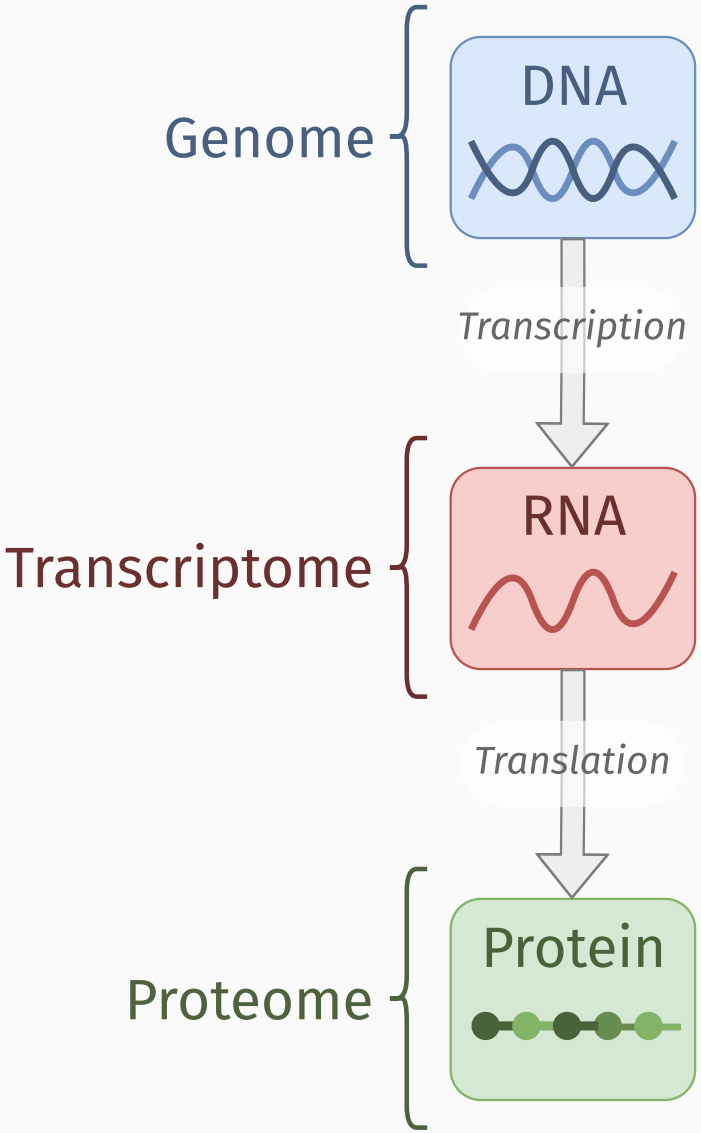
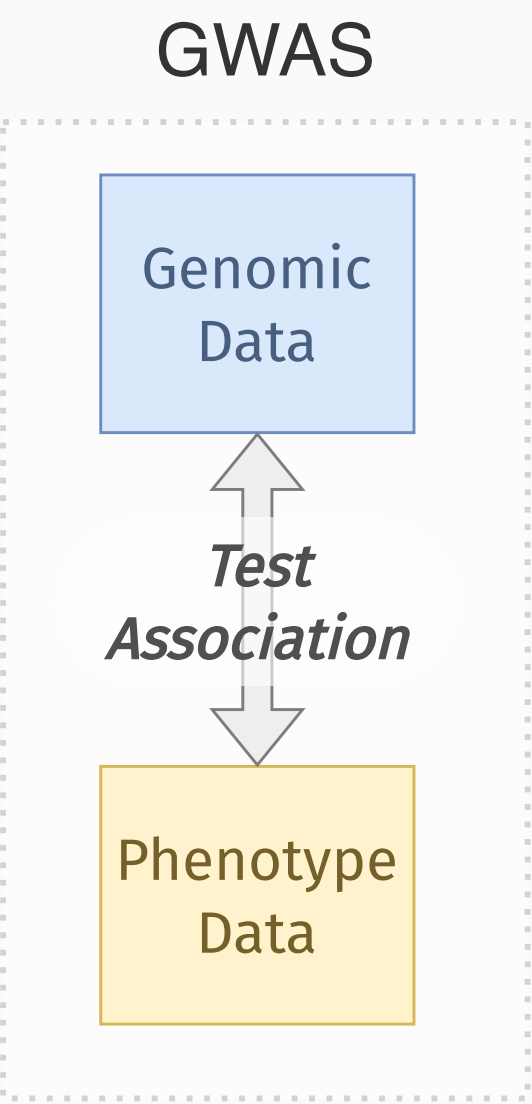
Background: GWAS, TWAS, & PWAS

Central Dogma of Molecular Biology



Genome-Wide Association Studies (GWAS)

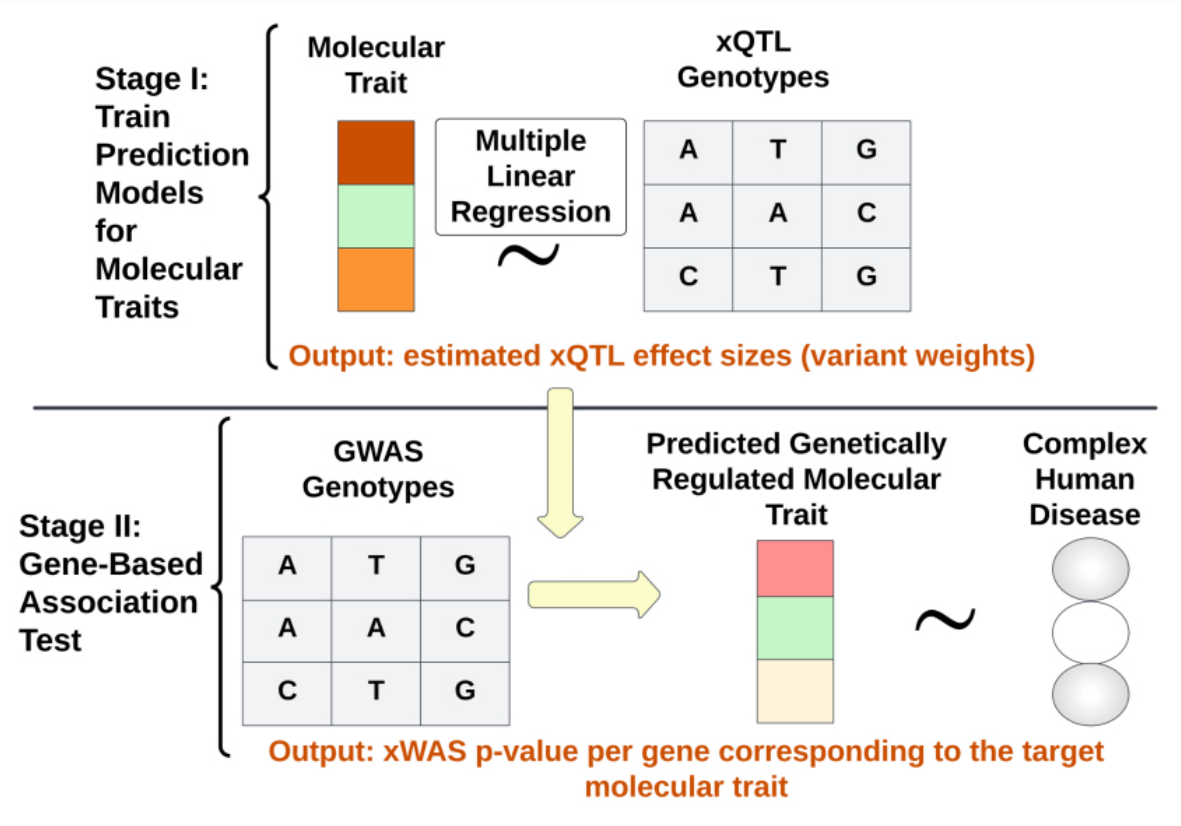


- identify genetic variants associated with a trait of interest (phenotype)
- most GWAS-identified variants are in non-coding regions[1–7]
- limited statistical power due to enormous number of variants tested, smaller effect sizes[2,8]
- which of the potentially highly-correlated SNPs in an identified is causal?[3]

  ⇒ issue with unknown biological mechanisms

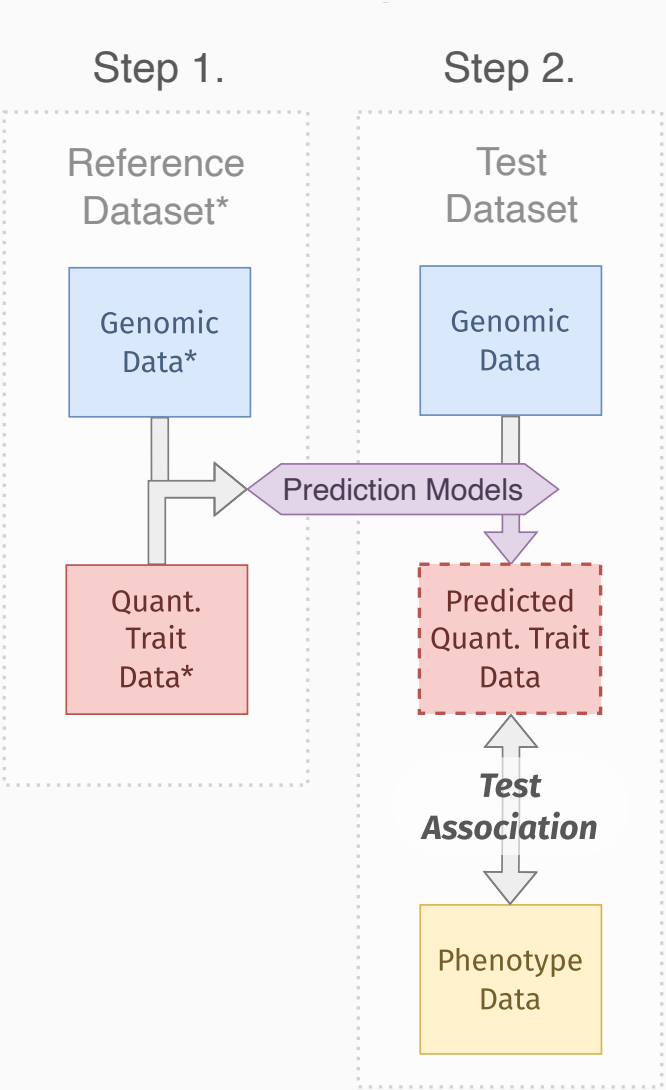
  - expression quantitative trait loci (eQTLs) – regulatory variants that modify transcript abundance (ie, gene expression)[9–14]
  - protein quantitative trait loci (pQTLs) – regulatory variants that modify protein abundance[15,16]

x-Wide Association Studies (xWAS)[15,16,20–24]



- TWAS/PWAS used to ID genes with genetically-regulated gene expression/protein abundance (GReX/GReP) associated with a phenotype[15,17–19]
  $\rightarrow$ illustrate biological mechanisms of GWAS risk variants

Standard Two-Stage xWAS[16,20–22]



1) Train models

train quantitative trait prediction model from reference genotype, $\mathbf{G}_0$, and quantitative trait (gene expression, protein abundance, etc.), $\mathbf{Q}_0$, data from the same subjects

$$\mathbf{Q}_0 = \mathbf{G}_0\mathbf{w} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim N(0,\ \sigma_\epsilon^2\mathbf{I})$$

where $\mathbf{w}$ is xQTL weights

2) Test association

test the association between predicted genetically-regulated quantitative trait (GReX or GReP) and phenotype of interest using estimated xQTL weights $\widehat{\mathbf{w}}$

$$\widehat{\mathbf{GReX}} = \mathbf{G}_{\text{test}}\widehat{\mathbf{w}} \qquad \text{(Individual-level)}$$

BGW-xWAS Tool

BGW-TWAS: Background

BGW-TWAS Paper

Bayesian Genome-wide TWAS Method to Leverage both *cis*- and *trans*-eQTL Information through Summary Statistics. 2020. *AJHG*. `https://pmc.ncbi.nlm.nih.gov/articles/PMC7536614/`.

- BGW-TWAS[25] tool originally developed for TWAS
- can be applied to other kinds of QTLs, not just eQTLs ("BGW-**x**WAS")
- **Motivation:**
  - existing TWAS methods only used *cis*-SNPs (variants within ±1MB of a gene) to fit GReX imputation models
  - *trans*-SNPs explain variation in expression quantitative traits, may be significant themselves

  ⇒ including *trans*- variants expected to increase imputation accuracy of GReX and the power of TWASs
- **Problem:** "the enormous computational cost required to fit 20K GReX imputation models for genomewide genes and **10M genotypes** per tissue type"

BGW-xWAS Steps

1) Obtain Summary Statistics

generate single variant xQTL summary statistics (aka Score Statistics)

2) Prune Genome Segments

Select only the top `${max_blocks}` (default: 50) *trans*-genome blocks to use in joint-model training.

3) Train Quantitative Trait Prediction Model

Train gene expression prediction model using single variant xQTL summary stats (obtained in Step 1) for genome blocks (obtained in Step 2)

~~4) Predict Bayesian GReX/GReP/etc. (optional)~~

~~Predict GReX for test genotype data~~

Used for individual-level test data only; later application instead uses summary-level test data.

5) Association Studies

Perform using predicted GReX/GReP/etc. (obtained in Step 4) or using summary-level data from GWAS.

BGW-xWAS: Model

Bayesian variable selection regression (BVSR) with both cis- and trans-eQTLs:

$$\mathbf{E}_g = \mathbf{G}_{cis}\mathbf{w}_{cis} + \mathbf{G}_{trans}\mathbf{w}_{trans} + \boldsymbol{\epsilon} \qquad \epsilon_i \sim N\left(0, \sigma_\epsilon^2\right)$$

$$w_{cis,i} \sim \pi_{cis}\, N\left(0, \sigma_{cis}^2\, \sigma_\epsilon^2\right) + \left(1 - \pi_{cis}\right) \partial_0\left(w_{cis,i}\right)$$

$$w_{trans,i} \sim \pi_{trans}\, N\left(0, \sigma_{trans}^2\, \sigma_\epsilon^2\right) + \left(1 - \pi_{trans}\right) \partial_0\left(w_{trans,i}\right)$$

- xQTL effect size w_i follows a "spike-and-slab" prior distribution–ie, we only retain a subset of the possible regressors
- $\Pr\left\{w_i \sim N\left(0, \sigma_q^2\sigma_\epsilon^2\right)\right\} = \pi_q$ vs. $\Pr\{w_i = 0\} = 1 - \pi_q, \quad q \in \{cis, trans\}$
- latent indicator vector $\boldsymbol{\gamma}_{p\times 1}$ is introduced to aid computation: $\gamma_i \sim \text{Bern}(\pi_q)$
- $\text{E}\{\gamma_i\}$ is the posterior probability (PP_i) for the ith SNP to be an eQTL with non-zero effect size w_i.

BGW-xWAS: Model

EM-MCMC Algorithm:
combines expectation–maximization (EM) algorithm with Markov Chain Monte Carlo (MCMC) algorithm

- **E-step**: Use MCMC per genome block to estimate $(\mathbf{w}, \mathbf{PP})$ from most recent estimation of $(\boldsymbol{\pi}, \boldsymbol{\sigma}^2)$
- **M-step**: update $(\boldsymbol{\pi}, \boldsymbol{\sigma}^2)$ using new estimates of $(\mathbf{w}, \mathbf{PP})$ from E-step

Iterate until convergence, parallelize over genome blocks

Final Estimate: $(\widehat{\mathbf{w}}, \widehat{\mathbf{PP}})$ **Predicted GReX/GReP/etc.**:

$$\widehat{\mathbf{GReX}} = \sum_{i=1}^{p} \widetilde{G}_i(\widehat{PP}_i\, \widehat{w}_i) \qquad \widetilde{\mathbf{G}} = \text{test genotype data} \qquad \text{(Individual-level xWAS)}$$

BGW-TWAS: Summary-Level Association Test

GWAS summary statistics are often publicly available, and can be used to conduct summary-level association tests without the need for individual-level test genotype/phenotype data

Can use burden Z-score as implemented by S-PrediXcan[17]

$$\widetilde{Z}_g = \sum_{i \in \text{Model}} \frac{(\widehat{PP}_i\, \widehat{w}_i)\, \widehat{\sigma}_i\, Z_{\text{GWAS},i}}{\sqrt{(\widehat{\mathbf{PP}}\, \widehat{\mathbf{w}})'\, \mathbf{V}\, (\widehat{\mathbf{PP}}\, \widehat{\mathbf{w}})}}, \qquad \widehat{\sigma}_i^2 = \text{Var}(\mathbf{G}_{0,i}), \qquad \mathbf{V} = \text{Cov}(\mathbf{G}_0)$$

$\widehat{PP}_i$ estimated posterior causal probability for SNP i
$\widehat{w}_i$ estimated xQTL effect-size for SNP i
$Z_{\text{GWAS},i}$ Z-score statistic values for single variant test (ie, GWAS)
$\widehat{\mathbf{PP}}$ estimated posterior causal probabilities from trained model
$\widehat{\mathbf{w}}$ xQTL effect-sizes from trained model
$\mathbf{V}$ genotype covariance matrix (LD data) estimated from reference panel genotype data $\mathbf{G}_0$
$\mathbf{G}_0$ reference genotype matrix
$\widehat{\sigma}_i$ estimated standard deviation of SNP i from genotype vector for SNP i: $\mathbf{G}_{0,i}$

Application TWAS & PWAS

Datasets

Training Data: ROS/MAP[26]

Religious Orders Study and Rush Memory and Aging Project. 2018. *J. Alzheimer's Dis.*.
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6380522/.

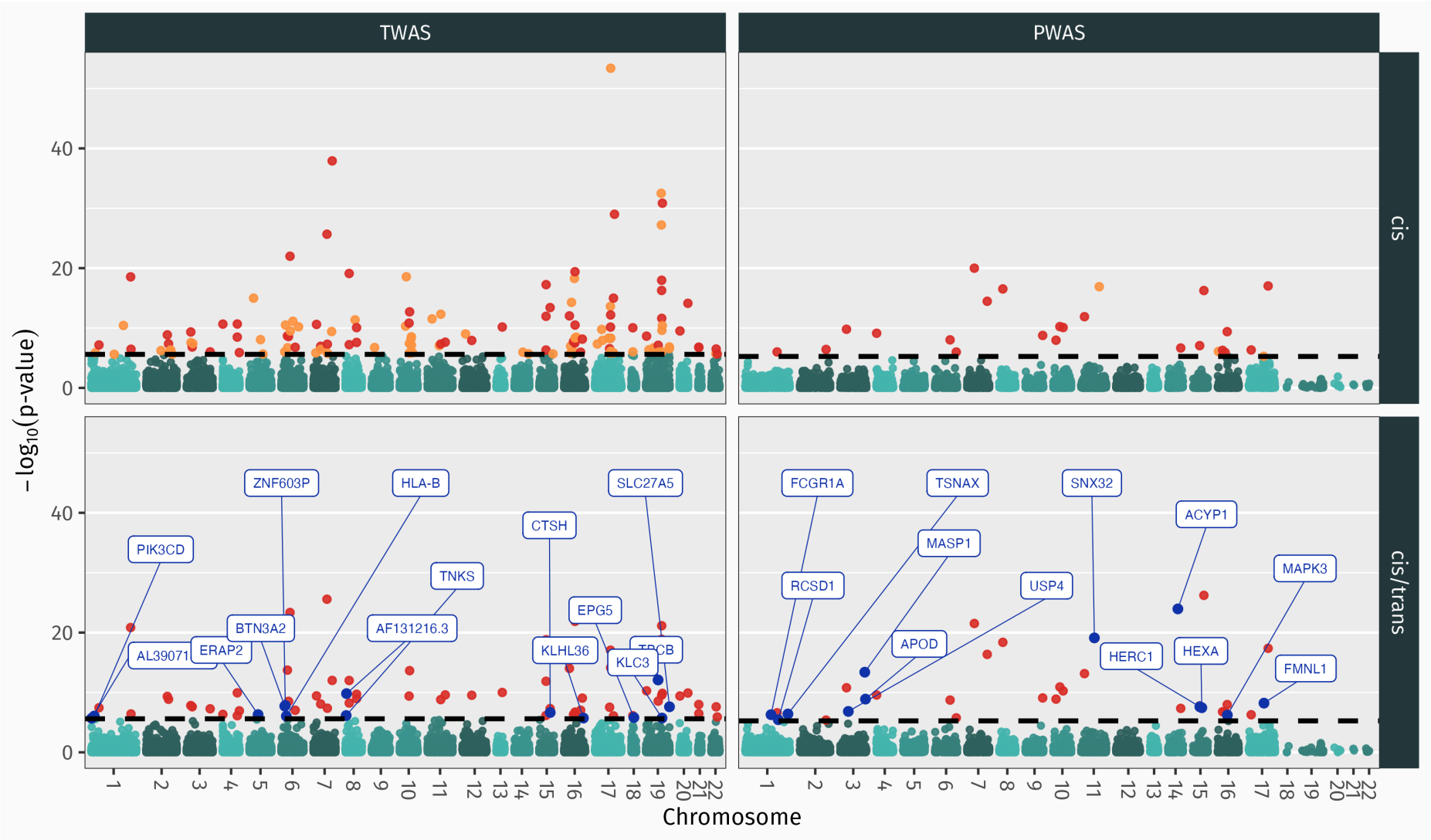
Dorsolateral prefrontal cortex (DLPFC) tissue data	N	Genes
RNA-Seq (Gene Expression)	931	21,244
TMT Proteomics (Protein Abundance)	716	12,175

GWAS of Alzheimer's Disease[27]

New insights into the genetic etiology of Alzheimer's disease and related dementias. 2022. *Nat. Genet.*.
https://www.nature.com/articles/s41588-022-01024-z.

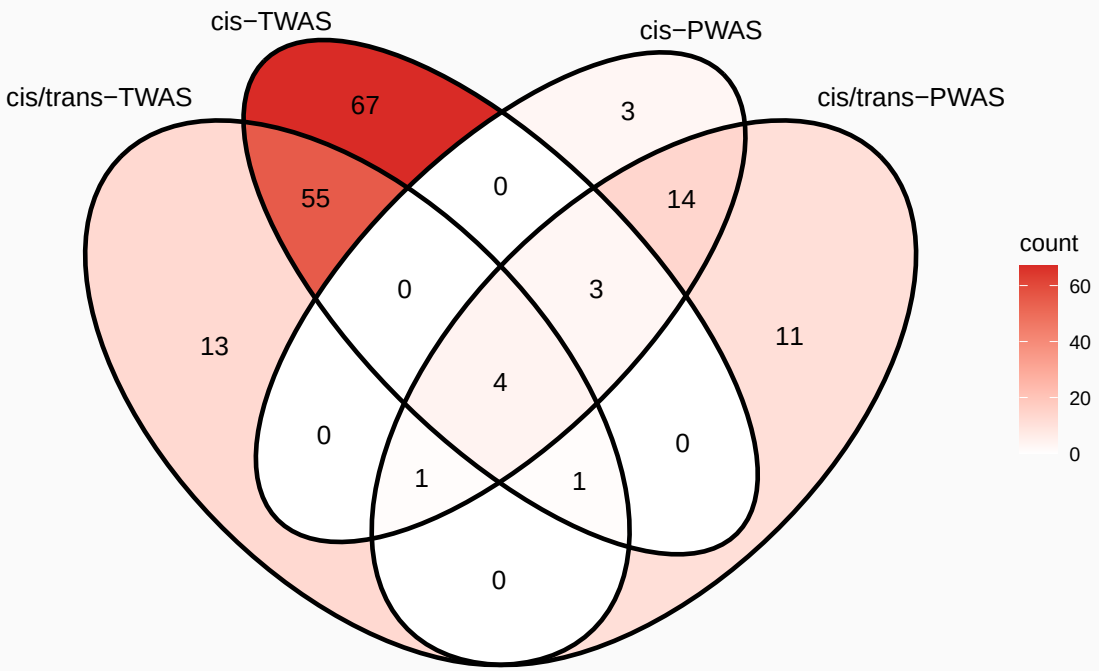
- 111,326 clinically diagnosed/"proxy" AD cases
- 677,663 controls

Manhattan Plots



Results

	Total Genes	Significant	Nonsignificant	NA	*trans*-only
cis-TWAS	16408	130	15973	305	0
cis/trans-TWAS	16676	74	16047	555	298
cis-PWAS	8829	25	7553	1251	0
cis/trans-PWAS	8922	34	7595	1293	84



Overlap of Significant Risk Genes

Results

Risk Genes Identified by all xWAS

CHR	Gene	Function/Disease Association
7	*EGFR*	epidermal growth factor receptor
8	*EPHX2*	lipid mediator; mutations associated with familial hypercholesterolemia
15	*LACTB*	regulator of mitochondrial lipid metabolism
17	*ACE*	angiotensin-converting enzyme

Gene Ontology Pathway Enrichment Analysis (pathDIP[28], PANTHER[29])

Enriched Paths Significant for all xWAS
Alzheimer disease-amyloid secretase
EGF receptor signaling
Insulin/IGF pathway-mitogen activated protein kinase kinase/MAP kinase cascade
Interferon-gamma signaling
Ras Pathway

Summary

Summary

- *trans*-SNPs are important for explaining variation in expression quantitative traits, may be significant themselves
- traditional xWAS tools only incorporate *cis*-SNPs
- BGW-xWAS tool accounts for both *cis*- and *trans*- variants
- BGW-xWAS uses various strategies make this computationally efficient: EM-MCMC, parallel computation, data pruning
- application study of AD xWAS identified risk genes previously identified in GWAS
- application study of AD xWAS show risk genes significantly enriched for AD-related pathways

Summary

Limitations:

- assumes a two-stage model for xWASs
- computational costs still substantial to train models for genome-wide genes per tissue type, per -omic data type:
- $\sim$30min of computation time per gene (in parallel with 4 CPU cores)
- PWAS based on protein abundance; does not consider effect of genetic variants on protein function[8]

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